BRIEF-003: Sub-50ms TTFT Under Extreme Concurrent Load

Latency Engineering & Event-Driven Inference 2026

STAGE-X.SPACE TECHNICAL ARCHITECTURE BRIEF
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EXECUTIVE SUMMARY:
Holding sub-50ms Time-To-First-Token on smart-grid state estimation across five million concurrent telemetry streams under zero-data-retention constraints.

KEY ARCHITECTURE PROOFS:
1. Network Layer: AF_XDP kernel-bypass networking with pinned CPU thread affinity.
2. Zero Retention: Ephemeral in-memory ring buffers cryptographically zeroed post-inference.
3. Speculative Decode: Pipelined draft verification for rapid anomaly classification.
4. Audit Logging: High-speed append-only syslog datagram output over Unix domain sockets.

FIELD MEASUREMENTS:
- P99 Time-To-First-Token: 44.8ms under 300% simulated load spike
- Concurrency: Sustained 120,000 operations per second per server node
- Memory Residue: 0 bytes retained post-session verification